

UI/UX Design Brief

Product: ExamGuard — Smart Exam Allocation & Identity Verification System **Document Version:** 1.0 **Date:** August 14, 2026

1. Design Principles

- **Clarity under pressure.** Exam day is high-stress for students and invigilators alike. Every screen should communicate status in under 2 seconds of glance time — large type, unambiguous color coding, minimal choices.
- **Zero-install access.** The student experience must feel instant and native in a browser: fast load, responsive layout, no forced app download, works on low-end Android devices and older iOS Safari versions.
- **Confidence through feedback.** Every action (scan, verify, sync) gives immediate, unmistakable visual and haptic/audio feedback, since invigilators are moving quickly through a queue of students.
- **Calm authority.** Visual language should feel institutional and trustworthy — restrained color palette, generous whitespace, no playful or casual UI patterns that would undercut the seriousness of identity verification.
- **Offline transparency.** The invigilator app always makes sync state visible (synced / pending / offline) so users never wonder whether a verification "counted."

2. Visual System

Element	Direction
Color palette	Neutral base (charcoal/white/gray) with a single institutional accent (deep blue). Status colors reserved exclusively for verification states: green (verified/correct), red (wrong hall/error), amber (pending/locked), gray (absent/inactive).
Typography	A single, highly legible sans-serif (Inter) — one weight for UI chrome, bold weight reserved for status and key data (hall, seat, name); a monospace face for roll numbers and codes.
Iconography	Simple line icons (camera, lock, checkmark, X, search) — no illustrative or decorative icons that compete with data.
Elevation & shape	Card-based layout with soft shadows and 8px corner radius for a modern, approachable but professional feel.
Motion	Minimal, purposeful: the T-30 unlock uses a brief unlock animation; the scan result uses a full-screen color flash (200ms) rather than a modal, so invigilators don't need to dismiss anything before scanning the next student.

2.1 Color Palette

The palette is deliberately restrained: a neutral base for structure and text, one institutional accent for brand and primary actions, and four reserved status colors that appear nowhere else in the UI. Reserving status colors exclusively for verification states is what makes the green/red scan feedback instantly legible — if amber or red ever appeared in ordinary chrome, invigilators would lose that half-second of unambiguous recognition.

Neutral Base			
	Ink #10131A	Primary text, headings	Highest-emphasis text on light surfaces
	Charcoal #1F2430	Body text	Default body copy
	Slate Gray #6B7280	Secondary text, disabled state	Captions, placeholders, inactive/absent status
	Border Gray #D7DEE8	Dividers, table borders, input outlines	Structural lines, not text
	Surface Tint #F4F7FB	Page background, striped rows	Low-contrast background fill
	White #FFFFFF	Card and surface background	Primary content surface

Institutional Accent			
	Deep Blue (Primary) #1A3C6E	Headers, primary buttons, links, active nav	The single brand color; used sparingly to keep it authoritative rather than decorative
	Deep Blue — Hover #14315C	Button/link hover and pressed states	~15% darker than primary
	Blue Tint #EAF0FB	Selected rows, info banners, focus rings	Background-only, never used for text

Status Colors (reserved — verification states only)			
	Verified Green #1E8E5A	Correct-hall scan flash, VERIFIED chips, success toasts	Tint #E6F6EE for backgrounds/badges
	Alert Red #D93025	Wrong-hall scan flash, WRONG_HALL / error states	Tint #FDEAE9 for backgrounds/badges
	Pending Amber #C77700	Locked hall/seat panel, ALREADY_USED / duplicate-scan warning, pending-sync	Tint #FFF4E0 for backgrounds/badges
	Inactive Gray #6B7280	ABSENT status, disabled controls, offline indicator	Tint #EEF0F3 for backgrounds/badges

Usage rules: status colors never appear in navigation, buttons, or chrome outside their specific verification context. The deep-blue accent is never used to signal success/error, so it can't be confused with a status cue. Every status color meets at least 4.5:1 contrast against white for text use, and its paired light tint is used only as a background behind dark text/icons, never as a text color itself.

2.2 Typography System

Primary typeface: Inter (variable weight), self-hosted or via Google Fonts, with a system-font fallback stack for maximum load speed on low-end devices: `Inter, -apple-system, "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-serif`.

Monospace typeface: IBM Plex Mono (fallback: `"Roboto Mono", "SFMono-Regular", Consolas, monospace`) — reserved for roll numbers, seat numbers, barcode tokens, and any other identifier where character-by-character clarity matters more than warmth.

Style	Typeface / Weight	Size	Line height	Usage
Display	Inter Extrabold (800)	28–32px	1.2	Reveal moment — hall & seat number on the student pass
H1	Inter Bold (700)	22–24px	1.3	Page/screen titles
H2	Inter Bold (700)	18px	1.35	Section headers, card titles
H3	Inter Semibold (600)	15px	1.4	Sub-section labels, list group headers
Body	Inter Regular (400)	14–15px	1.55	Default body copy, form fields, table cells
Body Strong	Inter Semibold (600)	14–15px	1.55	Candidate name, key data on the verification card
Caption	Inter Regular (400)	12px	1.4	Timestamps, helper text, footnotes
Label / Chip	Inter Semibold (600), uppercase, 0.04em tracking	11px	1.2	Status chips (VERIFIED, ABSENT, etc.), form labels
Mono	IBM Plex Mono Medium (500)	14–16px	1.4	Roll numbers, seat codes, barcode token display

Typography rules: only two families are used system-wide (Inter for everything except identifiers, Plex Mono for identifiers) to keep the interface calm under exam-day stress. Bold weight is reserved for genuinely important data — status, names, hall/seat — never for routine UI chrome, so emphasis stays meaningful. Minimum body size across all three surfaces is 14px (12px only for captions) to remain legible at arm's length on the invigilator's phone and in bright hall entrances.

3. Admin Web Console

Primary users: exam-cell staff, working at a desktop, often under deadline pressure during setup week.

Key screens: - **Dashboard (home):** exam-cycle overview — upcoming exams, halls configured, students mapped, any validation errors awaiting resolution. - **Roster Upload:** drag-and-drop CSV/Excel with a live preview table, inline validation errors highlighted per row, and a "Fix and re-upload" loop rather than an all-or-nothing failure. - **Hall Management:** table/grid view of halls with capacity utilization bars per exam. - **Exam & Mapping Builder:** a stepper flow — (1) define exam details, (2) select roster segment, (3) auto-allocate or manually assign hall/seat, (4) review and confirm — with a visual seat-map grid for manual overrides. - **Reveal Settings:** simple form with a clear preview ("Students will see their hall at 9:30 AM for a 10:00 AM exam") to avoid off-by-one timing confusion. - **Live Attendance Dashboard:** hall cards showing verified/pending/absent/wrong-hall counts with color-coded progress rings, auto-updating without refresh; drill-down to a per-student log. - **Audit Log & Export:** filterable table with CSV/PDF export, timestamps in institution-local time with UTC on hover.

4. Student Web Portal

Primary users: students, overwhelmingly on mobile browsers, often checking their pass in low-signal areas outside exam halls.

Key screens: - **Login:** roll number + password, large touch targets, clear forgot-password path. - **Timetable:** card list of upcoming exams sorted chronologically, each showing course, date, time, and a status chip (Upcoming / Pass Ready / Completed). - **Exam Pass (core screen):** - Above the fold: course name, date/time, student photo and name for self-verification, and the dynamic barcode rendered large and high-contrast for easy scanning even at arm's length. - Hall/seat panel: before T-30, shows a locked-padlock state with a countdown ("Unlocks in 1h 12m"); at T-30, this panel animates open to reveal building, room, and seat number in large bold type, with a small building-directions link if geodata is available. - A persistent note explaining that the barcode refreshes automatically and should not be screenshotted, since screenshots will not scan successfully once expired. - **Empty/edge states:** no-exams-mapped state, expired-session re-login prompt, and a clear "already verified" confirmation state shown after a successful scan (detected via a lightweight status poll) so students know they're checked in.

5. Invigilator Mobile Scanner App (Expo)

Primary users: invigilators standing at a hall entrance, scanning continuously for the first 15–20 minutes of an exam window, often one-handed while holding a clipboard.

Key screens: - **Home / Hall Selector:** on launch, invigilator selects (or auto-loads, if pre-assigned) their hall and exam; shows pre-sync status ("Roster ready — 142 students cached") before scanning can begin. - **Scanner (primary, default screen):** full-screen camera viewfinder with a scan-target frame; minimal chrome so the camera occupies maximum area for fast, reliable scans in varied lighting. - **Verification Result overlay:** - Correct hall: full-screen green flash, large candidate photo, name, roll number, seat number, and a single "Verify" confirmation button (or auto-confirm after a configurable delay for maximum speed). - Wrong hall: full-screen red flash, same candidate card, plus a prominent "Correct Hall: Block B, Room 214" redirect instruction. - Already used: distinct amber/red state with "Already Verified at 9:41 AM" to catch duplicate/shared-pass attempts immediately. - **Manual Search:** a single search field with instant roll-number autocomplete, surfacing the same verification card as a successful scan, for damaged passes or dead phones. - **Sync Status Bar:** persistent, unobtrusive bar at the top or bottom of every screen showing "12 pending sync" / "All synced" / "Offline mode" so invigilators always know their data safety state without leaving the scanner. - **Session Summary:** end-of-exam screen totaling verified/absent/flagged counts for the invigilator's hall, with a one-tap "Submit to Admin" action once online.

6. Accessibility & Responsiveness

- Student portal targets WCAG 2.1 AA: minimum 4.5:1 text contrast, full keyboard navigability, screen-reader labels on the barcode and lock/unlock states.
- All three surfaces support both light and dark system themes, defaulting to light for print/legibility in bright exam-hall entrances.
- Admin console is desktop-first but remains usable on tablets for on-site adjustments during exam week.
- Invigilator app is designed exclusively for one-handed, portrait, mobile use — no landscape or tablet layout planned for v1.

7. Content & Tone

- Copy is short, direct, and instructional ("Scan the student's barcode," "Hall details unlock in 12 minutes") — no marketing tone anywhere in the product.
- Error states always pair a plain-language explanation with a next action ("This pass has already been used. If this is a mistake, use Manual Search.").
- Status labels are consistent system-wide: VERIFIED, WRONG_HALL, FLAGGED, ABSENT — the same vocabulary appears in the UI, the API, and the exported audit logs, so support conversations never have to translate between "screen language" and "log language."